

Appendix

Role of Fine Aggregate in Airfield Paving Concrete Mixture Designs

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Table A-1. Physical properties of aggregates.

Aggregate Type	Nominal Maximum Size, mm (in)	ASTM C33 Gradation	Flatness and elongated ratio (3:1) (%)
Limestone	25 (1.0)	#57	14
Limestone	9.5 (0.371)	#8	9
Granite	25 (1.0)	#57	13
Granite	9.5 (0.371)	#78/ #8	11
Natural Sand 1	4.75 (#4)	n/a	n/a
Natural Sand 2	4.75 (#4)	n/a	n/a
Manufactured Sand	4.75 (#4)	n/a	n/a

Table A-2. Morphological properties of aggregates.

Property	Natural Sand 1	Manufactured Sand
<i>Fineness Modulus</i>	2.68	3.36
<i>Specific Gravity</i>	2.61	2.76
<i>% Passing No.200</i>	0.03	0.02
<i>Fines Content (%)</i>	0.01	0.02
<i>Geology</i>	Silica / Quartz	Dolomitic siltstone, Reagan sandstone, glauconitic sandstone
<i>Uncompacted Voids Content (%)</i>	39%	49%
<i>AIMS Angularity Index (mean)</i>	2420 ± 99	3280 ± 77
<i>AIMS Form Index (mean)</i>	6.41 ± 0.17	8.35 ± 0.10
<i>Observed Shape Classification</i>	Well-rounded	Very angular

Table A-3. Statistical Summary of Compressive Strength.

Laboratory	Curing Age	Fine Sand Range	Significant Drop-Off Threshold
Lab 1	7 Days	None identified	N/A
Lab 1	28 Days	None identified	N/A
Lab 2	7 Days	21.0% – 35.0%	38.50%
Lab 2	28 Days	21.0% – 35.0%	38.50%

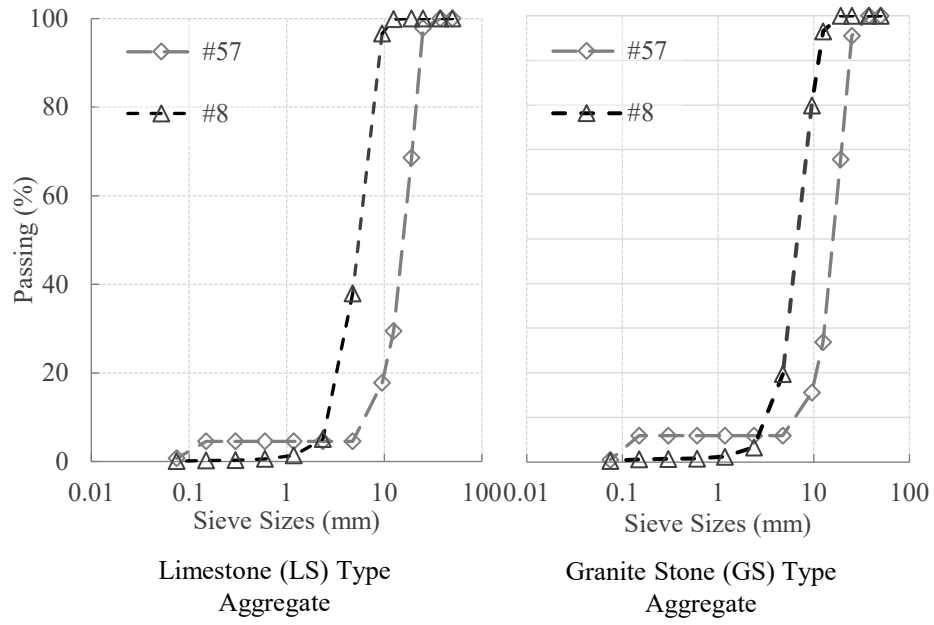


Fig. A-1. Sieve analysis of coarse and intermediate sized aggregates.

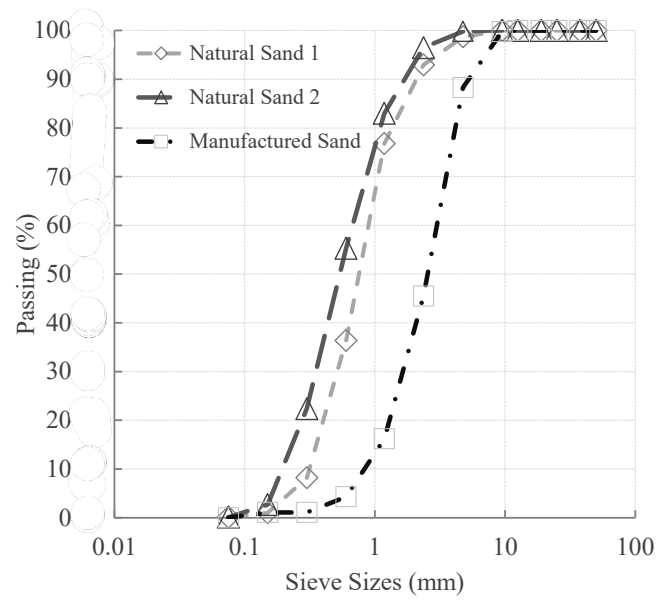


Fig. A-2. Particle size distribution of fine aggregates.

		
5	4	3
Collapse	Surface Voids > 50%	Surface Voids 30%-50%
		
	2	1
	Surface Voids 10%-30%	Surface Voids < 10%

Fig. A-3. Surface void rating scale (1-4). A higher rating scale indicates a higher amount of surface voids and poor response to consolidation. A collapse of concrete sample is rated 5, acquired from Uddin 2024.

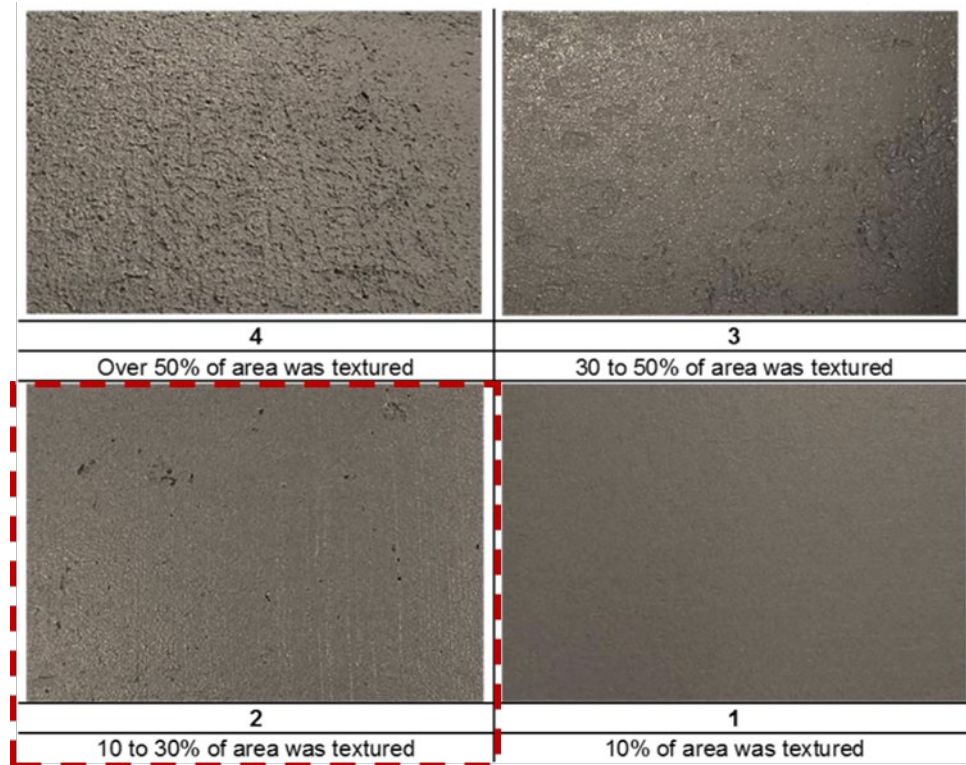


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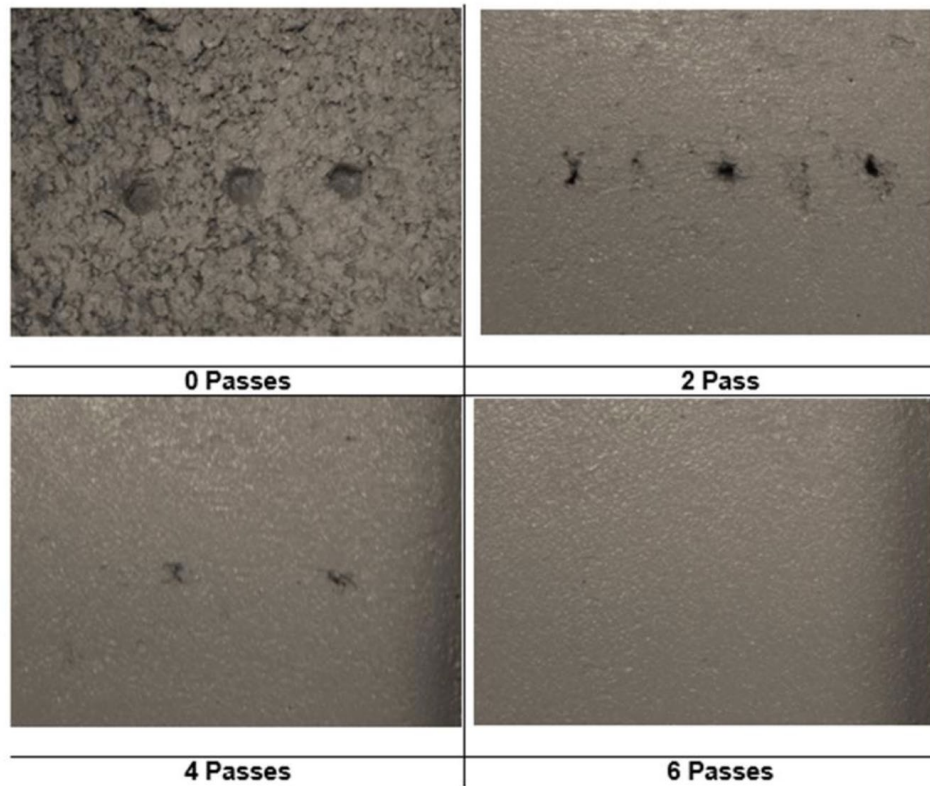


Fig. A-5. An example of the number of passes needed to close surface holes (Image acquired from Cook 2015).

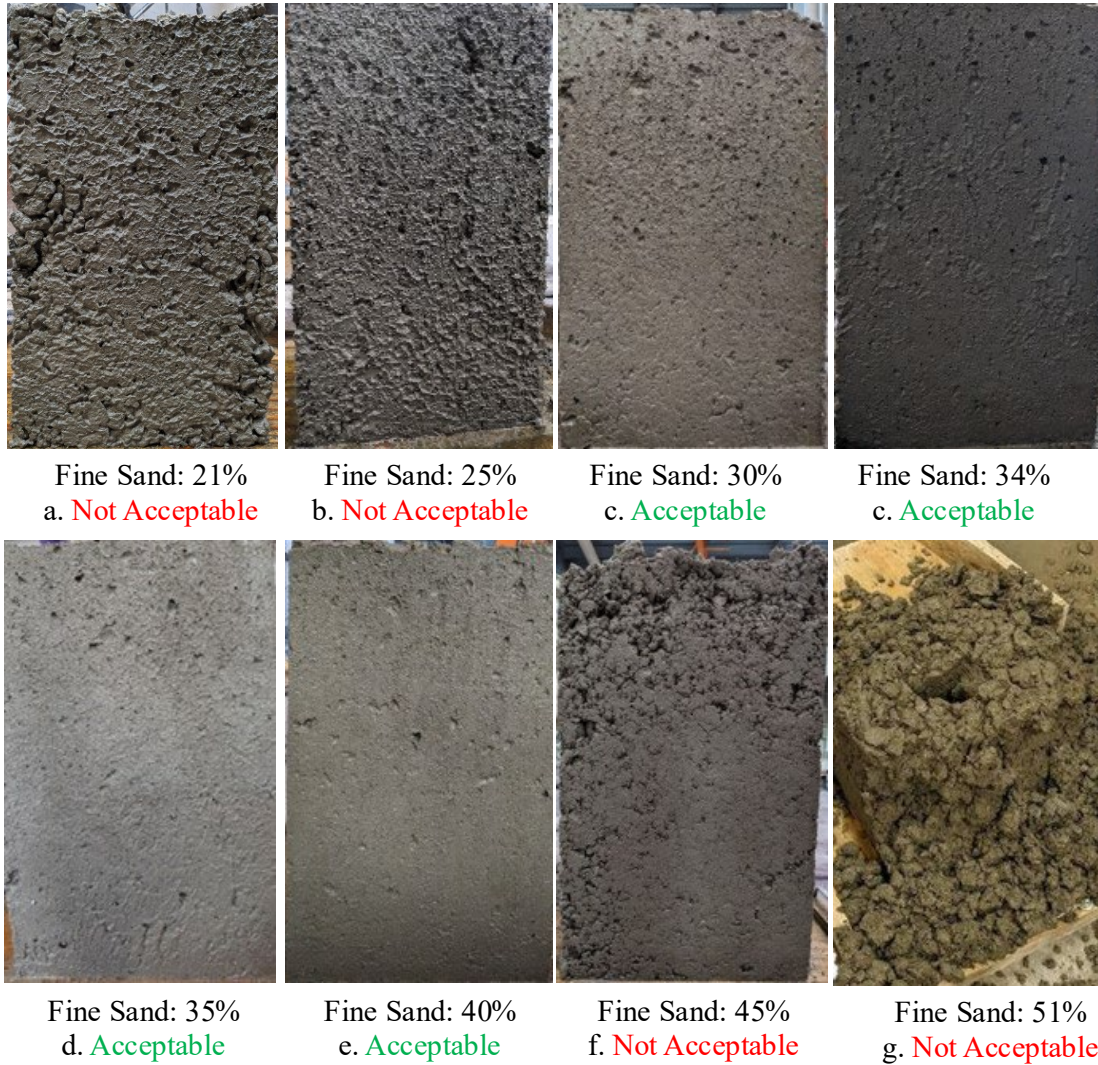


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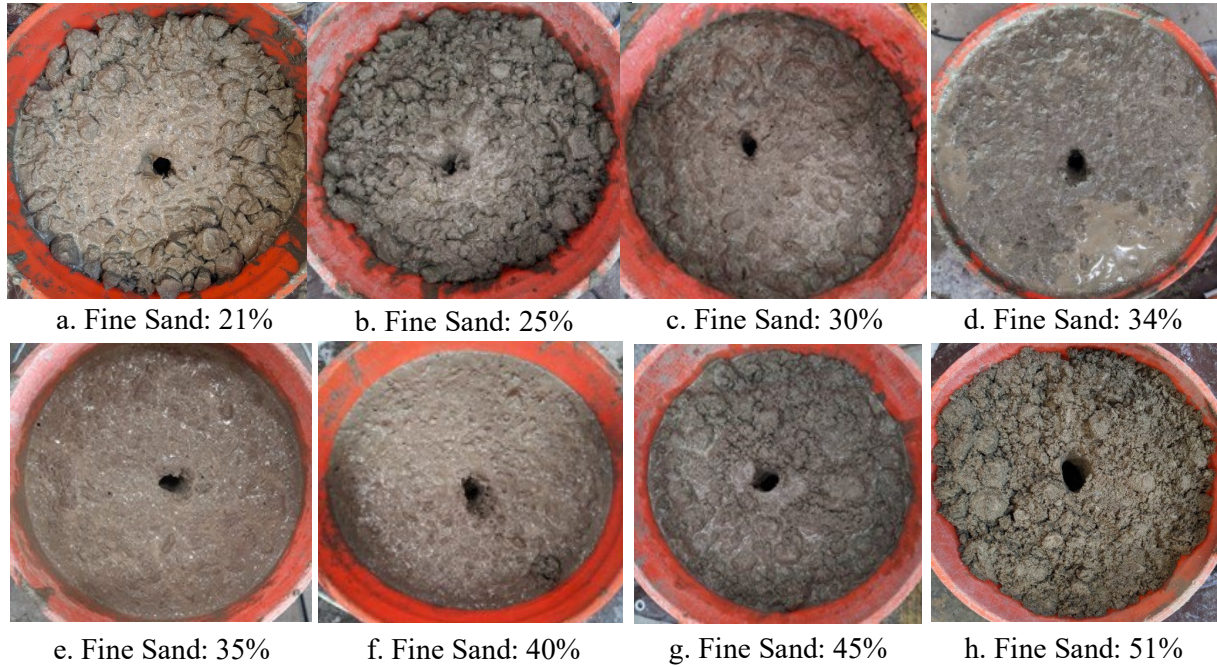


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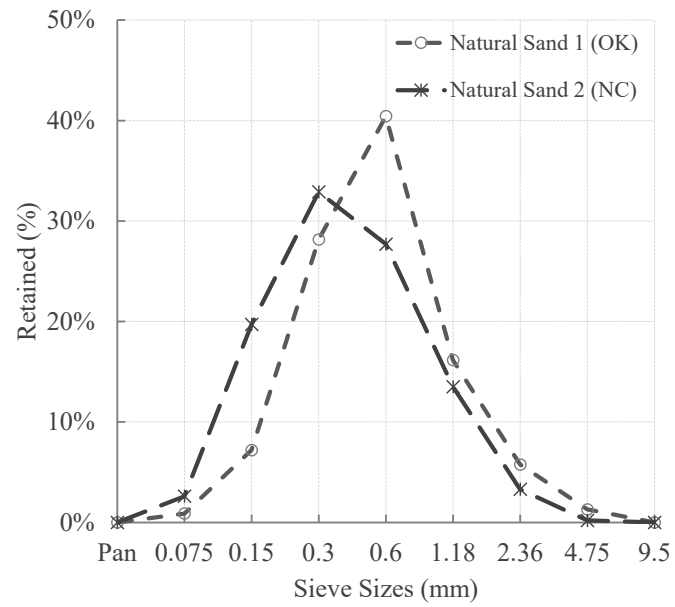


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